



Statistics 101: Margin of Error in Polling — A Full Data-Analysis Session

Statistics • Statistics

COURSE-OBJ.STAT101.2

Interpret margin of error and confidence intervals correctly, and determine whether an apparent change between two polls reflects a real shift or ordinary sampling variation. A complete 90-minute session packet with warm-up, guided practice, partner/lab work, independent practice, extension work, assessment, and a full answer key.

PRINT • TEACH • REVISIT



Plan the session

Course and objective

Statistics · Statistics; COURSE-OBJ.STAT101.2 — Interpret a poll's margin of error and confidence interval, and determine whether two poll results are statistically distinguishable or within expected sampling variation.

What is included

Cover; instructor guide; key-term and concept preview; original reading or worked examples; warm-up (2 items); guided practice with organizer (4 items); partner/lab discussion (3 items); independent practice (6 items); small-group extension work (4 items); assessment; complete answer key; access and sources page.

Suggested pacing (90-minute session)

Warm-up 8 min · read both poll case files + vocabulary 15 min · guided interval-calculation organizer 18 min · partner comparison discussion 18 min · independent practice 22 min · extension work 5 min · assessment 4 min.

Preparation

Print student pages separately from the answer key, or distribute through a secure LMS. Review the reading/examples and organizer before class. No paid textbook or app is required.

Student directions

Read or work through the material carefully. Support every answer with evidence, work shown, or clear reasoning.



Before you begin

Key terms

sample — a subset of a population surveyed to estimate the whole

margin of error — the range of likely error around a poll estimate

confidence interval — estimate $\pm$ margin of error, at a stated confidence level

sampling bias — a flawed method that skews who gets surveyed

statistically distinguishable — two results whose intervals do not substantially overlap

sampling variation — natural result-to-result differences from chance alone

Concept focus

A poll surveys a sample, not the whole population, so its result carries sampling error. A 95% confidence interval (candidate support $\pm$ margin of error) means: if the same polling method were repeated many times, about 95% of the resulting intervals would contain the true population value. Margin of error shrinks as sample size grows, roughly proportional to $1/\sqrt{n}$, so quadrupling the sample only halves the margin of error. Two poll results are statistically distinguishable only when their confidence intervals do not substantially overlap; a shift smaller than the combined margin of error may simply be sampling noise, not a real change in opinion.

Model 1 — Reading a single poll result

A poll of 1,000 voters shows Candidate A at 52%, margin of error ± 3 points. The true population support is estimated to fall between 49% and 55% about 95% of the time this method is used — not "exactly 52%."

Model 2 — Comparing two polls for a real change

Poll 1: Candidate A at 50% (± 3). Poll 2, two weeks later: Candidate A at 53% (± 3). The intervals (47-53) and (50-56) overlap substantially, so this 3-point shift is within expected sampling variation, not strong evidence of a real change.

WARM-UP

Warm-up

Answer using what you already know or recall from the previous session.

1. What does " ± 3.5 percentage points at a 95% confidence level" mean, using the lesson's own definition?

2. Calculate Reyes's full confidence interval in Poll A, showing your addition and subtraction.



Poll A: The City Council Race, Six Weeks Out

A local newspaper commissions a poll of the city council race between incumbent Dana Reyes and challenger Marcus Webb. The pollster randomly selects 800 registered voters likely to vote, using a mix of phone and text-message outreach to reduce the chance that any one group (for example, only landline users, who skew older) is overrepresented. The result: Reyes 47%, Webb 44%, undecided 9%, with a stated margin of error of ± 3.5 percentage points at a 95% confidence level.

The newspaper's headline reads "Reyes Leads Webb by 3 Points," but the paper's data editor pushes back before publication. Reyes's true support could reasonably be anywhere from about 43.5% to 50.5%, and Webb's from about 40.5% to 47.5% — the two ranges overlap substantially, roughly from 43.5% to 47.5%. That overlap means the poll cannot rule out the possibility that Webb is actually ahead, or that the race is exactly tied. The editor revises the headline to "Reyes Narrowly Ahead of Webb, Race Within Margin of Error" instead, a smaller but more accurate claim than "leads."



Poll B: A Follow-Up Poll Two Weeks Later

Two weeks before the election, the same newspaper commissions a second poll, this time surveying 1,600 likely voters — double the first poll's sample size — using the same phone-and-text methodology. The result: Reyes 49%, Webb 42%, undecided 9%, with a stated margin of error of ± 2.5 percentage points.

A reporter drafts a story claiming "Reyes's Lead Has Widened From 3 Points to 7 Points," treating the shift from Poll A to Poll B as clear evidence that Reyes gained ground. The data editor again intervenes: doubling the sample size reduced the margin of error, but not proportionally — from ± 3.5 to ± 2.5 , not to ± 1.75 , because margin of error shrinks with the square root of sample size, not sample size directly. More importantly, Reyes's confidence interval in Poll B (46.5% to 51.5%) barely overlaps Poll A's interval (43.5% to 50.5%) at the very top, while Webb's interval in Poll B (39.5% to 44.5%) does not overlap Poll A's interval (40.5% to 47.5%) by much either. The editor concludes this pattern is more consistent with a real shift toward Reyes than the first comparison would have been, but still asks the reporter to describe the change as "an apparent gain for Reyes, though within the range sampling variation alone could produce" rather than stating it as certain.



Given / Evidence

Conclusion

1. Calculate Webb's full confidence interval in Poll A.
2. Why did the data editor reject the original "Reyes Leads Webb by 3 Points" headline for Poll A?
3. Explain why doubling the sample size from 800 to 1,600 did not cut the margin of error in half.
4. Calculate Reyes's and Webb's confidence intervals in Poll B, showing your work.

Partner / lab discussion

Discuss each item with a partner or lab group before writing a joint response.

1. With a partner, compare Reyes's intervals across both polls (43.5-50.5 in Poll A; 46.5-51.5 in Poll B). Explain in your own words why the data editor sees this as weak evidence of a real gain rather than strong proof.

2. With a partner, discuss: why does the lesson describe the phone-and-text outreach method as reducing the chance that "any one group is overrepresented"? What could go wrong with a phone-only poll?

3. Discuss: the reporter wanted to say Reyes's "lead widened from 3 points to 7 points." Explain what is misleading about treating the raw point totals ($47-44=3$ vs. $49-42=7$) as the full story.



Independent practice

Work independently. Show your work or cite evidence in every answer.

1. Why does the lesson say a poll surveys "a sample, not the whole population"? What is the practical consequence of this for any single poll number?

2. A classmate says "a 1,600-person poll is always better than an 800-person poll." Evaluate this claim using evidence from the lesson.

3. Explain, using the definition of "statistically distinguishable" from the lesson, why Poll 1 vs. Poll 2 in the Model 2 example ($50\% \pm 3$ vs. $53\% \pm 3$) was called "not strong evidence of a real change."



Independent practice

Work independently. Show your work or cite evidence in every answer.

4. Design an original two-poll scenario (with made-up but plausible numbers and margins of error) where the shift between polls IS statistically distinguishable. Explain why using interval math.

5. Design an original two-poll scenario where the shift is NOT statistically distinguishable, and explain why.

6. Explain what "undecided" voters (9% in both polls) mean for interpreting the race, beyond the margin-of-error math.



Small-group extension work

Use for review or additional application practice.

1. Write a two-to-three sentence, accurate headline for Poll B's results that a careful data editor would approve, using the lesson's cautious framing as a model.

2. Explain why a pollster reports a margin of error at all, instead of just reporting the single percentage (e.g., "47%") as the answer.



Small-group extension work

Use for learners ready for additional challenge or synthesis.

1. Compare the two case files: what specific methodological choice stayed the same between Poll A and Poll B, and what changed? Why does this matter for comparing them fairly?

2. Write a short paragraph (4-5 sentences) explaining to a non-statistics friend why "Reyes leads by 3 points" and "the race is within the margin of error" can both be defensible ways to describe the exact same Poll A data.

Assessment

Complete independently to show what you learned today.

1. Self-assessment: which is harder for you right now — calculating a confidence interval, or explaining in words why two overlapping intervals are not strong evidence of a real change? Name one specific thing you would review.



Assessment · self-assessment

Rate today's work:

4 — I can teach this to someone else.

3 — I can do this independently.

2 — I can do this with help.

1 — I need to revisit this before the next session.



Answer key

1. If the same polling method were repeated many times, about 95% of the resulting margin-of-error intervals would contain the true population value; it does not mean the reported number is exactly correct.
2. $47\% - 3.5\% = 43.5\%$; $47\% + 3.5\% = 50.5\%$. Reyes's interval is 43.5% to 50.5%.
3. $44\% - 3.5\% = 40.5\%$; $44\% + 3.5\% = 47.5\%$. Webb's interval is 40.5% to 47.5%.
4. Because Reyes's and Webb's confidence intervals overlap substantially (roughly 43.5% to 47.5%), the poll cannot rule out Webb actually being ahead or the race being tied, so "leads" overstates the certainty the data supports.
5. Margin of error is roughly proportional to $1/\sqrt{n}$ (one over the square root of the sample size), not $1/n$; since $\sqrt{2}$ is about 1.41, not 2, doubling the sample size only reduces the margin of error by a factor of about 1.41, not 2 — consistent with the drop from ± 3.5 to ± 2.5 rather than to ± 1.75 .

Accept equivalent, well-supported alternative answers throughout.



Answer key

6. Reyes: $49\% - 2.5\% = 46.5\%$; $49\% + 2.5\% = 51.5\%$, so 46.5% to 51.5%. Webb: $42\% - 2.5\% = 39.5\%$; $42\% + 2.5\% = 44.5\%$, so 39.5% to 44.5%.

7. The two intervals only barely overlap at the top (around 46.5-50.5), which is more consistent with a real shift than a large overlap would be, but the overlap is not zero, so ordinary sampling variation between two separately drawn samples still cannot be fully ruled out as the explanation.

8. A phone-only poll (especially landline-only) could overrepresent groups more likely to have and answer landlines, such as older voters, producing a sample that does not accurately reflect the full population of likely voters — a form of sampling bias distinct from margin of error.

9. The raw point gap ignores the margin of error on each poll; a gap can grow numerically between two polls without that growth being statistically meaningful, since each individual poll result already carries several points of uncertainty in either direction.

10. Because interviewing every voter is impractical, pollsters interview a subset and use it to estimate the whole population's views; the practical consequence is that any single reported number is an estimate with built-in uncertainty (the margin of error), not an exact measurement.

Accept equivalent, well-supported alternative answers throughout.



Answer key

11. A larger sample does reduce margin of error (as shown going from Poll A to Poll B), so it is generally more precise, but the claim is incomplete: sample size alone does not fix sampling bias from a flawed method (e.g., a non-representative outreach method), so a larger biased sample can still be worse than a smaller, well-designed one.

12. The two confidence intervals (47-53 and 50-56) overlap substantially, so the 3-point shift falls well within what ordinary sampling variation between two separate samples could produce, meaning the poll cannot confidently distinguish a real opinion shift from noise.

13. Answers vary; must show two confidence intervals that do not overlap (or barely overlap), with the numbers/margins explicitly stated and the non-overlap explained as evidence the shift is unlikely to be pure sampling noise.

14. Answers vary; must show two confidence intervals that substantially overlap, with the shift explained as plausibly explainable by sampling variation alone.

15. Undecided voters represent voters whose eventual choice is not yet captured by either candidate's percentage; with a 5-point-or-smaller gap between candidates and 9% undecided, the undecided bloc alone could be large enough to decide the race regardless of the current margin-of-error analysis.

Accept equivalent, well-supported alternative answers throughout.



Answer key

16. Answers vary; must avoid overstating certainty (no "clear lead" or "widening lead" claimed as fact) and should reflect that the shift is an apparent, not confirmed, change, similar to "Reyes Shows Modest Gain Over Webb, Though Race Remains Close" or similar appropriately hedged language.

17. Reporting only the raw percentage without the margin of error would falsely imply the number is exact; the margin of error communicates the real uncertainty inherent in estimating a full population's views from a sample, which is essential for readers to interpret the result correctly.

18. The phone-and-text outreach methodology stayed the same; the sample size changed (800 to 1,600). Keeping the method constant while changing sample size makes the comparison fairer, since a change in results is less likely to be explained by a change in methodology rather than a real or sampling-driven shift.

19. Answers vary; must explain that the raw numbers do show Reyes ahead by 3 points, but the margin of error means the true difference could be zero or even reversed, so a cautious/complete description accounts for both facts rather than only reporting the headline number.

20. Answers vary; must name a specific skill from the lesson (interval calculation or interval-overlap interpretation) and one concrete thing to review.

Accept equivalent, well-supported alternative answers throughout.



Access, sources, and license

Extension

Ask for a second method, a counterexample, or a short student-led extension of the concept, with a brief explanation of the choice.

Accommodated seating/504

Follow the individual accommodation plan: extended time, breaks, enlarged print, reduced copying, or an alternate response method as documented.

Additional support

Pre-teach key vocabulary or notation before class. Model the first guided item aloud. Accept oral or dictated responses when appropriate.

Multilingual learners

Preview key terms with examples. Allow brief first-language discussion before independent writing. Provide sentence or solution starters.

Course-level alignment · accessed September 18, 2026

COURSE-OBJ.STAT101.2 — Interpret a poll's margin of error and confidence interval, and determine whether two poll results are statistically distinguishable or within expected sampling variation. The objective is original wording describing a focused part of the course-level goal.

Course-level learning objective written for this packet; no accreditation, transfer credit, or universal course-equivalency is claimed.

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